AGENTIC-ARBITER -- FREE-COOLING DECISION REPORT

Data centres over-cool, continuously, because nobody can promise them the hours ahead. A chiller plant needs hours of notice to change mode, and a thermometer only ever reports NOW -- so the mechanical chillers keep running through hours that outside air could have cooled for free. FortyGuard closes that gap with heat intelligence 2 m above the ground, the height a ground-mounted condenser actually breathes. This agent turns that forecast into an hour-by-hour schedule carrying a calibrated safety bound.

THE SITE

Location	NY facility NY_way_1109643936 -- station KHZY, America/New_York
Building	OSM 1109643936 OSM way 1109643936
Facade gap	not applicable -- one building, no second facade
Weather record	43,036 real hourly records from KHZY
Plume physics	NOT MODELLED. No other tagged data centre lies inside the solver's validated range, so there is no neighbour intake for a plume to arrive at: the quantity does not exist here rather than being unmeasured. This is a statement about the model's domain, not a claim that recirculation here is zero. A building's own exhaust re-entering its own intake is not modelled at any site in this project.

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day	2025-03-14 (crossing)
Plant limit	21.0 C
Notice required	3 h
Forecast skill	0.50 relative to persistence
Level anchor	sensor
Condenser bank	longest facade
Switch budget	2 mode changes per day
Minimum dwell	3 h
Max dew point	15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 14 of 24 hours with 2 mode changes. 3 hours were safe but still ran chillers, because the schedule could not afford them.

Agent	14 of 24 hours free cooling, 2 mode change(s), 1 unsafe hour(s)
Incumbent	14 of 24 hours free cooling, 2 mode change(s), 1 unsafe hour(s) 3 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	FREE	yes	-	0.840	21.0	0.560	0.792
01	FREE	yes	-	0.865	21.0	0.560	0.726
02	FREE	yes	-	0.255	21.0	-1.110	0.889
03	FREE	yes	-	0.280	21.0	-1.110	0.892
04	FREE	yes	-	-0.000	21.0	-1.670	0.866

05	FREE	yes	-	-0.000	21.0	0.000	0.845
06	FREE	yes	-	1.110	21.0	2.220	0.716
07	FREE	yes	-	2.775	21.0	4.440	1.103
08	FREE	yes	-	6.390	21.0	6.670	2.017
09	FREE	yes	-	10.000	21.0	10.000	2.207
10	FREE	yes	-	13.050	21.0	13.330	2.276
11	FREE	yes	-	15.585	21.0	17.780	1.726
12	FREE	yes	-	18.060	21.0	20.560	1.485
13	FREE	yes	-	20.275	21.0	22.780	1.290
14	mechanical	no	dry-bulb	22.500	21.0	23.890	1.054
15	mechanical	no	dry-bulb	23.340	21.0	23.890	0.865
16	mechanical	no	dry-bulb	24.170	21.0	23.890	0.918
17	mechanical	no	dry-bulb	24.165	21.0	23.330	0.989
18	mechanical	no	dry-bulb	23.330	21.0	22.220	1.069
19	mechanical	no	dry-bulb	21.665	21.0	19.440	1.156
20	mechanical	no	dry-bulb	21.110	21.0	18.890	1.450
21	mechanical	yes	switch budget	20.555	21.0	18.890	1.419
22	mechanical	yes	switch budget	18.610	21.0	17.780	1.121
23	mechanical	yes	switch budget	18.060	21.0	16.670	1.024

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 0.840 C, which is 20.160 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.792 C, and the margin is measured, not chosen: 0.792 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

01:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 0.865 C, which is 20.135 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.726 C, and the margin is measured, not chosen: 0.726 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

02:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 0.255 C, which is 20.745 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.889 C, and the margin is measured, not chosen: 0.889 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

03:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 0.280 C, which is 20.720 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.892 C, and the margin is measured, not chosen: 0.892 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

04:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is -0.000 C, which is 21.000 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.866 C, and the margin is measured, not chosen: 0.866 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

05:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is -0.000 C, which is 21.000 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.845 C, and the margin is measured, not chosen: 0.845 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

06:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 1.110 C, which is 19.890 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.716 C, and the margin is measured, not chosen: 0.716 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

07:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 2.775 C, which is 18.225 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.104 C, and the margin is measured, not chosen: 1.104 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

08:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 6.390 C, which is 14.610 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 2.017 C, and the margin is measured, not chosen: 2.017 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

09:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 10.000 C, which is 11.000 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 2.207 C, and the margin is measured, not chosen: 2.207 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

10:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 13.050 C, which is 7.950 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 2.276 C, and the margin is measured, not chosen: 2.276 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

11:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 15.585 C, which is 5.415 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.726 C, and the margin is measured, not chosen: 1.726 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

12:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 18.060 C, which is 2.940 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.485 C, and the margin is measured, not chosen: 1.485 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

13:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 20.275 C, which is 0.725 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.290 C, and the margin is measured, not chosen: 1.290 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time. *** THIS WAS WRONG: the intake actually reached 22.780 C, above the limit. The bound failed here, and it is counted as a breach rather than explained away. ***

14:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.500 C against a 21.0 C limit, so it fails by 1.500 C. It would take a limit 1.500 C higher, or a bound 1.500 C tighter, to change this hour.

15:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 23.340 C against a 21.0 C limit, so it fails by 2.340 C. It would take a limit 2.340 C higher, or a bound 2.340 C tighter, to change this hour.

16:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 24.170 C against a 21.0 C limit, so it fails by 3.170 C. It would take a limit 3.170 C higher, or a bound 3.170 C tighter, to change this hour.

17:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 24.165 C against a 21.0 C limit, so it fails by 3.165 C. It would take a limit 3.165 C higher, or a bound 3.165 C tighter, to change this hour.

18:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 23.330 C against a 21.0 C limit, so it fails by 2.330 C. It would take a limit 2.330 C higher, or a bound 2.330 C tighter, to change this hour.

19:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.665 C against a 21.0 C limit, so it fails by 0.665 C. It would take a limit 0.665 C higher, or a bound 0.665 C tighter, to change this hour.

20:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.110 C against a 21.0 C limit, so it fails by 0.110 C. It would take a limit 0.110 C higher, or a bound 0.110 C tighter, to change this hour.

21:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 20.555 C against a 21.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

22:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 18.610 C against a 21.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

23:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 18.060 C against a 21.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

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